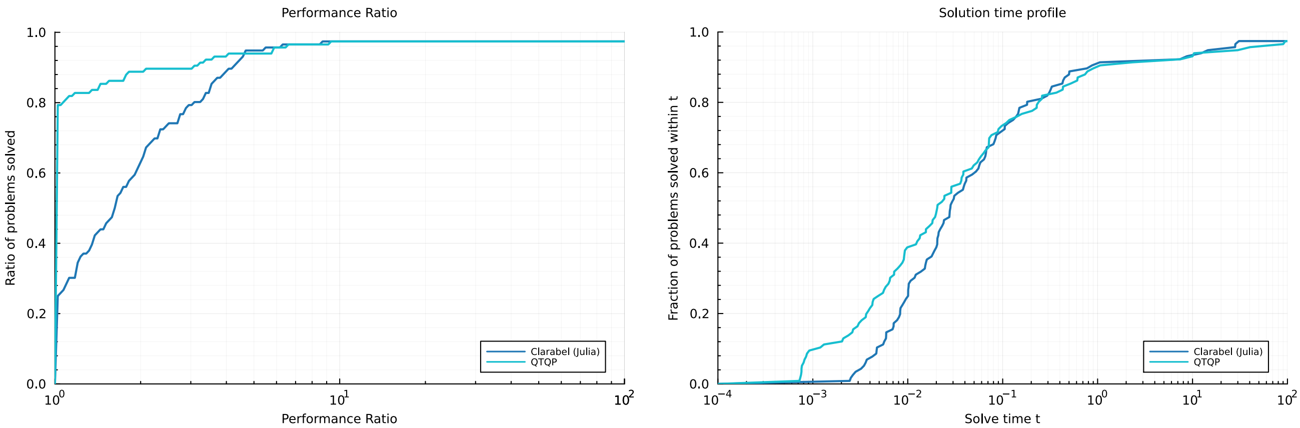
5 Numerical Experiments

5.1 Benchmark problems with quadratic objectives

5.1.1 The Maros-Meszaros test set

Figure 1: Performance profiles for the Maros-Meszaros problem set



(a) Relative performance profile

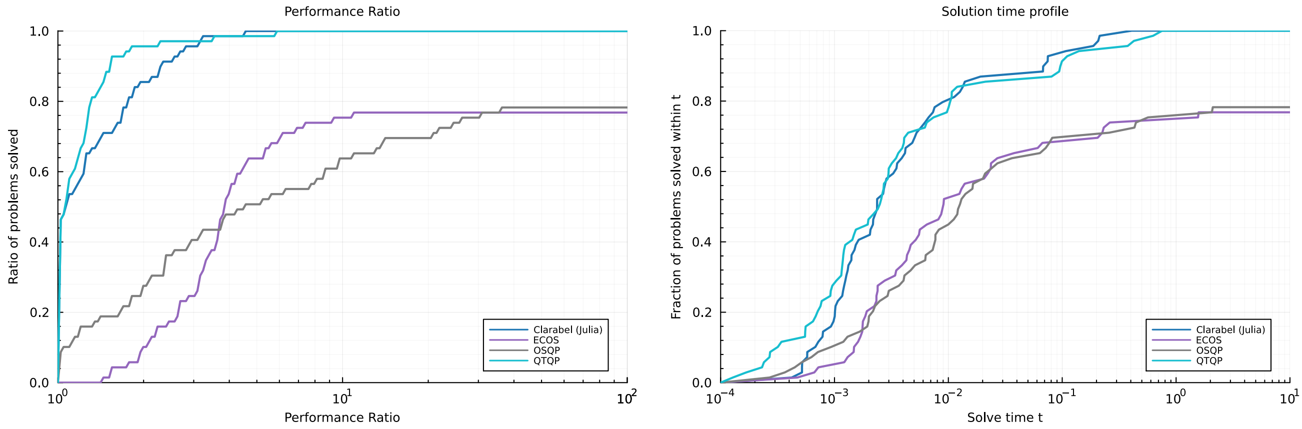
(b) Absolute performance profile

		Clarabel	**QTQP**
Shifted GM	Full Acc.	1.0	1.12
	Low Acc.	1.0	1.78
Failure Rate (%)	Full Acc.	2.6	2.6
	Low Acc.	0.0	2.6

(c) Benchmark timings as shifted geometric mean and failure rates

5.1.2 Constrained optimal control

Figure 2: Performance profiles for the optimal control problem set



(a) Relative performance profile

(b) Absolute performance profile

		Clarabel	**ECOS**	**OSQP**	**QTQP**
Shifted GM	Full Acc.	1.0	135.27	123.6	1.7
	Low Acc.	1.0	8.59	99.48	1.7
Failure Rate (%)	Full Acc.	0.0	23.2	21.7	0.0
	Low Acc.	0.0	1.4	18.8	0.0

(c) Benchmark timings as shifted geometric mean and failure rates

A Detailed benchmark results

Table 1: Solve times and iteration counts for the Maros-Meszaros problem set

Problem	vars.	cons.	nnz(A)	nnz(P)	iterations	time per iteration(s)	total time (s)

Table 2: Solve times and iteration counts for the Optimal Control problem set

					iterations	time per iteration(s)	total time (s)
Problem	vars.	cons.	nnz(A)	nnz(P)	ECOS	ECOS	ECOS
pendulum_3	28	23	83	23	11	4.29e-05	0.000471
forcesExample_1	38	29	83	28	14	4.75e-05	0.000666
forcesExample_2	38	29	83	27	13	9.38e-05	0.00122
forcesExample_3	42	32	92	31	12	6.02e-05	0.000722
toyExample_1	42	32	102	31	20	7.37e-05	0.00147
helicopter_1	48	30	96	39	20	7.43e-05	0.00149
nonlinearCstr_3	64	54	204	30	19	6.86e-05	0.0013
pendulum_1	78	63	243	63	12	0.000148	0.00178
pendulum_2	78	63	243	60	14	0.000126	0.00176
forcesExample_4	82	62	182	61	15	0.000117	0.00175
toyExample_2	82	62	202	61	18	0.000107	0.00192
aircraft_3	104	84	364	20	16	0.000213	0.00341
doubleInvertedPendulum_1	104	84	364	50	17	0.000141	0.00239
doubleInvertedPendulum_2	104	84	364	50	16	0.000173	0.00276
helicopter_3	118	86	278	80	22	0.000195	0.00428
aircraft_1	144	84	404	20	13	0.000143	0.00186
aircraft_2	144	84	404	40	13	0.00013	0.00169
aircraft_4	144	84	404	20	13	0.00018	0.00233
doubleInvertedPendulum_3	144	84	404	50	21	0.000166	0.00349
ballOnPlate_1	152	77	257	47	15	0.000156	0.00234
ballOnPlate_2	152	77	257	47	14	0.000115	0.00161
ballOnPlate_3	152	77	257	47	17	0.000141	0.00239
dcMotor_1	144	94	424	32	26	0.000314	0.00817
aircraft_10	164	104	444	20	13	0.000172	0.00224
aircraft_11	164	104	444	64	15	0.000303	0.00454
aircraft_12	164	104	444	64	15	0.00026	0.0039
ballOnPlate_4	252	127	427	77	19	0.000295	0.00561
helicopter_2	218	166	538	175	22	0.000294	0.00646
shell_1	249	159	559	60	15	0.000345	0.00517
shell_3	249	159	559	60	20	0.000217	0.00433
dcMotor_6	284	184	844	62	76	0.00031	0.0236
spacecraft_1	367	187	807	110	18	0.000307	0.00552
spacecraft_2	367	187	807	110	18	0.00026	0.00467
binaryDistillationColumn_1	311	266	2666	90	18	0.00048	0.00865
binaryDistillationColumn_2	311	266	2666	90	18	0.000773	0.0139
quadcopter_1	382	292	1162	110	16	0.000572	0.00915
quadcopter_6	382	292	1162	172	17	0.000533	0.00907
shell_2	489	309	1109	120	15	0.00059	0.00885
springMass_2	566	286	1646	181	20	0.000625	0.0125
springMass_3	566	286	1646	166	21	0.000621	0.013
tripleInvertedPendulum_3	636	366	2346	201	30	0.000686	0.0206
aircraft_13	804	504	2204	304	23	0.00117	0.0269
quadcopter_2	752	572	2312	220	18	0.0021	0.0378
quadcopter_4	752	572	2312	220	19	0.00117	0.0222
springMass_4	1126	566	3286	341	20	0.00118	0.0237

Table 2: Solve times and iteration counts for the Optimal Control problem set

					iterations	time per iteration(s)	total time (s)
Problem	vars.	cons.	nnz(A)	nnz(P)	ECOS	ECOS	ECOS
dcMotor_4	1404	904	4204	302	31	0.00216	0.067
quadcopter_3	1862	1412	5762	550	20	0.00305	0.0609
nonlinearChain_13	2457	2397	30857	660	22	0.0093	0.205
nonlinearChain_3	2457	2397	30857	660	22	0.0102	0.225
nonlinearChain_14	2637	2397	31037	660	24	0.0109	0.261
nonlinearChain_4	2637	2397	31037	660	24	0.00965	0.232
nonlinearChain_12	10377	9417	123977	2640	34	0.0459	1.56
nonlinearChain_2	10377	9417	123977	2640	34	0.0464	1.58